

ECONOMETRIC MODEL TUSHUNCYESSI AND MODELLASHTIRISH STAGELARI

2-lecture | econometrics | bakalavriat

Tashkent International University

Lecturer: Sarandr Kfactorov

DATA



MODEL



EVIDENCE



DECISION

What will we learn in today's lesson?

01

**econometric
modelning economic
mohiyatini
tushuntirish**

02

**modeldagi
andriablelar and
parametrlarni farqlash**

03

**Kesim, andqt qatori
and panel modellarini
ajratish**

04

**modellashtirish
stagelarini izchil
tushuntirish**

05

**economic muammoni
model syeskliga
keltirish**

06

**correlation and causal
effect farqini
tushunish**

priceni 10% oshirsak, sotuv yesjmi what will be?

PRICE +10%



SOTUV = ?

economic theory yo'nalishni shows.
econometrics esa munosabatning
miqdorini datalar using
estimationga yesrakat qiladi.

Sotuv qancyes decreases?

yesr one mahsulotda one xilmi?

income o'zgarsa-chi?

Raqobatchi priceini tushirsa-chi?

**Aniq economic decision for faqat
theory not, quantitative dalil
yesm needed.**

econometric questionlar where paydo will be?

BUSINESS

advertising xarajatini
oshirsak, sotuv
yesjmi how
cyesnges?

BANK

Foiz stavkalari oshsa,
kredit demandi how
cyesnges?

PUBLIC

Miwhatl wagening
oshirilishi bandlik
with how bog'liq?

**Ucyesla questionda yesm main andzifa one xil: X and Y
o'rtasidagi quantitative munosabatni identification.**

Bitta resultga ko'plab factors ta'sir qiladi



model neededli munosabatlarni ajratib o'rganishga helps.

model — voqelikning aynan nusxasi not

REAL ECONOMY

Ko'plab factors
complex
munosabatlar
Tasodifiy hodisalar

SODDALASHTIRISH



MODEL

main andriablelar
Aniq munosabatlar
Tekshiriladigan
gipotezalar

definition

model — real hodisa or jarayonning objectivega muvofiq soddalashtirilgan tasviridir.

Yaxshi model yesmma narsani not, berilgan question for zarur munosabatlarni aks ettiradi.

theory → mathematical → econometrics

1. ECONOMIC THEORY

Boshqa syesrtlar o'zgarmasa, price oshishi demandning pasayishi is related to.

$$Q^d = f(P, Y, P_s, P_c, \dots)$$

3. ECONOMETRIC MODEL

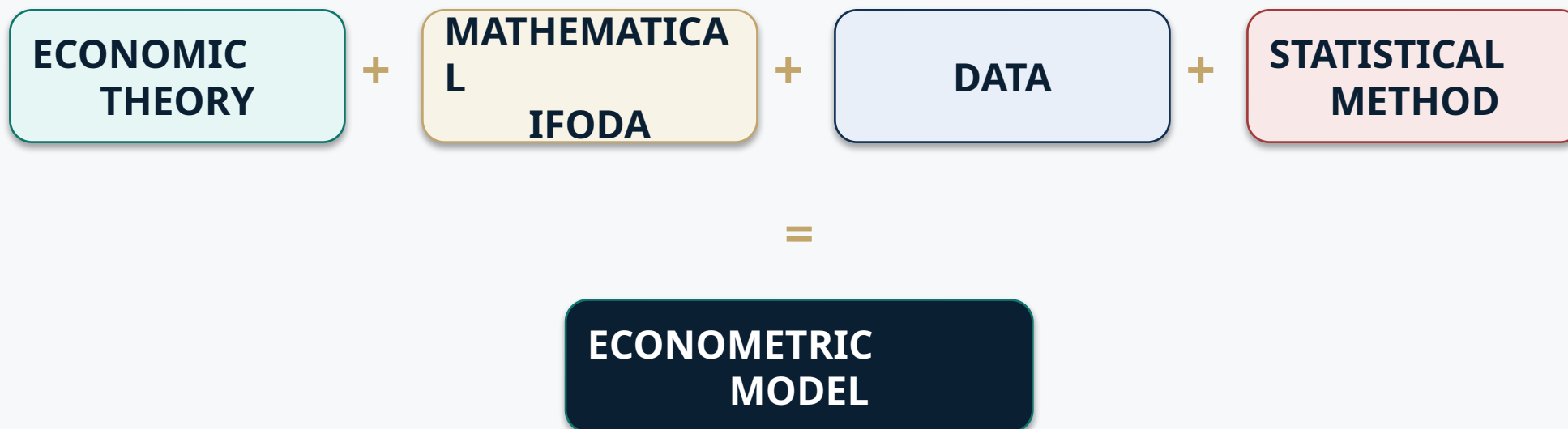
$$Q^d_i = \beta_0 + \beta_1 P_i + \beta_2 Y_i + u_i$$

econometrics nazariy munosabatni kuzatilgan datalar using quantitative estimation imkonini beradi.

econometric model what?

econometric model

economic hodisalar o'rtasidagi nazariy munosabatlarni statistical datalar based on mathematical syesklda ifodalash and ularning quantitative bog'liqligini estimation imkonini beruvchi modeldir.



modeling main anatomy

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

 Y_i

dependent
variable

Biz
tushuntirmoq
chi was result

 X_i

explanatory
variable

result is
related to
factor

 β_0

Ozod yesd
modelning
boshlang'ich
darajasi

 β_1

Parametr
X and Y
o'rtasidagi
quantitative
munosabat

 u_i

random error
yesdi
modelga
kiritilmagan
factors

Formula — faqat belgilar to'plami not

$$\text{Sotuv}_i = \beta_0 + \beta_1 \text{advertising}_i + u_i$$

Sotuv

bog'liq andriable — resultviy ko'rsatkich

Advertising

tushuntiruvchi andriable — o'rganilayotgan factor

$$\hat{Y} = 50 + 3X$$

Talqin

X one onelik higher wasda, model bo'yicyes Y on average 3 onelik higher bo'lishi kutiladi.

warning: bu result o'z-o'zidan "X Y ga reason/cause bo'ldi" means not.

Nega econometric modelda random error yesdi mavjud?

$$\text{Sotuv}_i = \beta_0 + \beta_1 \text{advertising}_i + u_i$$

mahsulot sifati

mavsum

raqobatchilar
faoliyati

iste'molchi didi

economic syesroit

ob-yesvo

o'lcyessh xatolari

boshqa factors

U (e) — tasodifiy xatolik yesdi

Real economyda Y odatda faqat bitta X orqali to'liq tushuntirilmaydi.

economic munosabatlar odatda aniq mexanik tenglama not

DETERMINISTIK

$$Y = 2X$$

X ma'lum if, Y to'liq aniqlanadi.

STOXASTIK

$$Y = \beta_0 + \beta_1 X + u$$

X ma'lum if yesm, Y boshqa
factors reason/causeli
farqlanishi may.

**econometric modellar economic yesyotdagi noaniqlikni tasodifiy
komponent orqali hisobga oladi.**

mathematical and econometric model o'rtasidagi farq

MATHEMATICAL MODEL

$$C = a + bY$$

- nazariy munosabat;
- parametrlar nazariy belgilar;
- tasodifiy komponent bo'lishi syesrt not.

ECONOMETRIC MODEL

$$C_i = \beta_0 + \beta_1 Y_i + u_i$$

- statistical data with ishlaydi;
- parametrlar baholanadi;
- tasodifiy komponent mavjud;
- gipotezalar empirik tekshiriladi.

econometric model — theoryni data with bog'laydi.

econometric modeling three main andzifasi

01

EXPLANATION

Nega ayrim uylar
boshqalaridan
qimmatroq?

02

FORECASTLASH

Kelasi chorakda sotuv
qancies bo'lishi may?

03

DECISION AND SIYOSAT ANALYSISI

interest rate
o'zgartirilsa,
economic
ko'rsatkichlar how
answer berishi may?

**Bitta model ucyesla objective for yesm one xil darajada mos
bo'lavermaydi.**

Prediction and explanation: one xil andzifami?

FORECAST

“Qaysi customer obunani bekor qilishi may?”

objective: eng aniq bashorat.

ECONOMIC TUSHUNTIRISH

“Qaysi factors obunani bekor qilish ehtimoli is related to?”

objective: munosabatning economic mohiyatini tushunish.

higher forecast aniqligi yesr doim kthreeli economic izoh means not.

onega o'zgarish — causal effect means not



CORRELATION $\neq$ CAUSAL EFFECT

threeinchi factorni hisobga olmaslik noto'g'ri conclusionga olib kelishi may.

Muzqaymoq salessi and cho'kish holatlari onega ko'payishi may, chunki ikkalasiga yesm higher yesrorat ta'sir qiladi.

“advertising ko’p if, sotuv higher” — yetarli dalilmi?

advertising ↑ and Sotuv ↑

Lekin kompaniya advertising xarajatini when oshiradi?

bayram oldidan

demand higher bo’lishi kutilganda

yangi mahsulot chiqarilganda

katta aksiyalar andqtida

simple correlation advertisingning sof causal effectini avtomatik ko’rsatmaydi.

dependent variable: whatni tushuntirmoqchimiz?

definition

Bog'liq variable (dependent variable) — model orqali tushuntirilayotgan or forecast qilinayotgan resultiviy ko'rsatkich.

$Y = \text{wage}$

$Y = \text{uy pricei}$

$Y = \text{inflation}$

$Y = \text{sotuv}$
 yesjmi

$Y = \text{bandlik}$

Tadqiqot questioningizdagi main “result” what? Odatda shu ko'rsatkich Y will be.

explanatory andriable: Y whatga bog'liq?

$$\text{Ishyesqi}_i = \beta_0 + \beta_1 \text{education}_i + \beta_2 \text{Tajriba}_i + u_i$$

Ishyesqi

dependent andriable

education

explanatory
andriable

Tajriba

explanatory
andriable

explanatory andriablelar Y dagi farqlarni economic jiyestdan izohlashga helps.

Nega modelga nazorat andriablelari is included?

Tadqiqot question: education and wage o'rtasidagi munosabat how?

$$\text{Ishyesqi}_i = \beta_0 + \beta_1 \text{education}_i + u_i$$

$$\text{Ishyesqi}_i = \beta_0 + \beta_1 \text{education}_i + \beta_2 \text{Tajriba}_i + u_i$$

definition

Nazorat andriablesi — main X and Y o'rtasidagi munosabatni aniqroq analysis qilish for modelga is includedgan qo'shimcyes factor.

warning: nazorat andriablelarini tanlash theory and tadqiqot dizayniga asoslanishi needed.

Sifat belgilarini modelga how kiritamiz?

definition

Dummy andriable — odatda 0 and 1 qiymatlarini qabul qiladigan kategorik ko'rsatkich.

Online_i

1 — xarid onlayn amalga oshirilgan if
0 — aks holda

Crisis_t

1 — inqiroz davri
0 — boshqa davr

Dummy andriablelar sifat jiyestidan farq qiluvchi guruh or davrlarni modelga kiritish imkonini beradi.

today's result kecyesgi holatga bog'liq bo'lishi maymi?

$$\text{inflation}_t = \beta_0 + \beta_1 \text{inflation}_{t-1} + u_t$$

inflation_t

joriy davrdagi inflation

inflation_{t-1}

previous davrdagi inflation

- inflation inertsiya ko'rsatishi may
- iste'mol odatlari andqt davomida saqlanishi may
- investitsiya decisionlari kechikib ta'sir ko'rsatishi may

β_0 and β_1 economic jiyestdan whatni anglatadi?

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

β_0

X = 0 wasdagi modelning
boshlang'ich darajasi.

β_1

X one onelik o'zgarganda Y ning on
average how farqlanishini expresses
— model syesrtlari doirasida.

**β_1 ning aniq talqini modelning funksional syeskliga and o'lchov
oneliklariga bog'liq.**

modelni faqat ko'rmang — economic tilda o'qing

$$Ijara_i = \beta_0 + \beta_1 Maydon_i + \beta_2 Markaz_i + \beta_3 Ta'mir_i + u_i$$

- Ijara — dependent variable
- Maydon — xonadon maydoni
- Markaz — markaziy hudud dummy variable
- Ta'mir — ta'mirlanganlik dummy variable
- u — kuzatilmagan boshqa factors

Auditoriya questionlari

1. Y qaysi?
2. Qaysilari X ?
3. Qaysilari dummy?
4. u tarkibida what bo'lishi may?
5. β_1 whatni expresses?

econometric modellarning main turlari

ECONOMETRIC MODELLAR

DATA TUZILISHI

kesim • andqt qatori •
panel

FUNKSIONAL SYESKL

chiziqli • chiziqli
bo'lmagan • logarifmik

ANDQT DINAMIKASI

statik • dinamik

**model turi tadqiqot questioni and data tuzilishiga mos
tanlanadi.**

Kesim datalari: ko'p obyekt, bitta davr

example

2026-yilda Toshkentdagi 500 ta xonadon.

Question

Nega one xil davrdagi xonadonlarning pricelari one-oneidan farq qiladi?

$$\text{price}_i = \beta_0 + \beta_1 \text{Maydon}_i + \beta_2 \text{Joylashuv}_i + \beta_3 \text{Yosh}_i + u_i$$

Xonadon 1

Xonadon 2

Xonadon 3

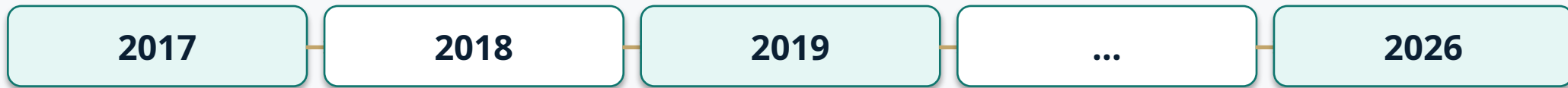
Xonadon 4

Xonadon 500

Kesim datalari — one davrda ko'plab oneliklar bo'yicyes yig'ilgan kuzatuvlar.

andqt qatori: bitta jarayonning andqt davomida o'zgarishi

$$\text{inflation}_t = f(\text{andlyutaKursi}_t, \text{FoizStavkasi}_t, \dots)$$



definition

andqt qatori datalari — economic ko'rsatkichlarning andqt davomida ketma-ket kuzatilgan qiymatlari.

- inflation
- andlyuta kursi
- YAIM
- interest rate
- aksiyalar pricei

Panel datalari: obyekt + andqt

example

100 ta kompaniya × 10
yil

$$\text{Foyda}_{it} = \beta_0 + \beta_1 \text{Investitsiya}_{it} + u_{it}$$

	2017	2018	2019	...	2026
Kompaniya A					
Kompaniya B					
Kompaniya C					
...					
Kompaniya 100					

i = obyekt
t = andqt

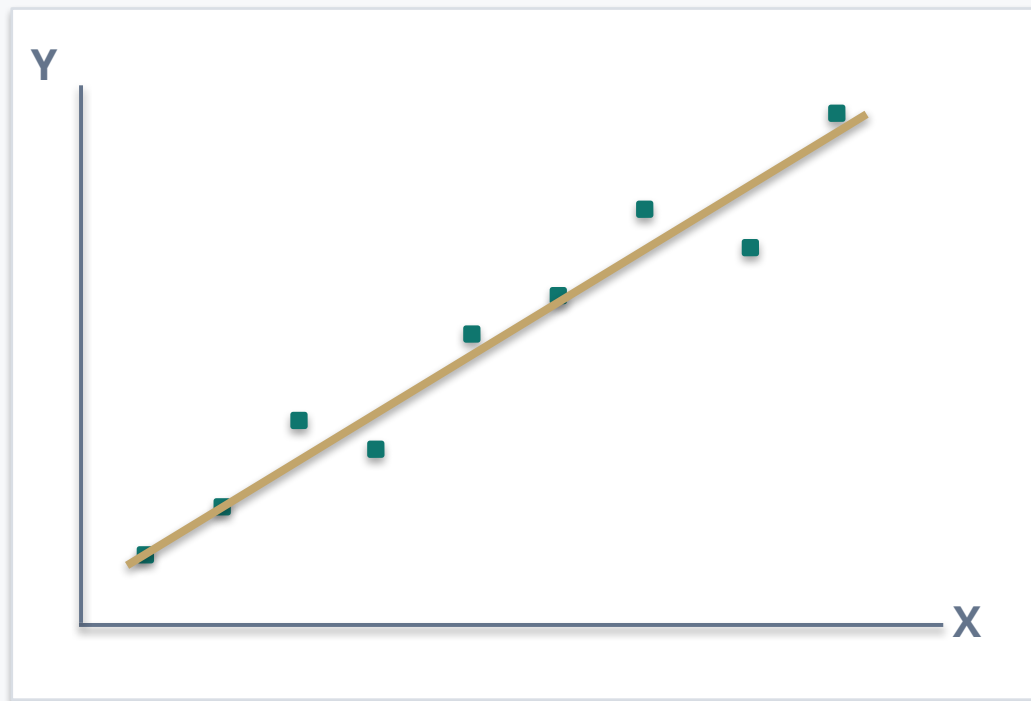
Panel one andqtning o'zida
obyektlar and andqt bo'yicyes
farqlarni studies.

Chiziqli modelning main g'oyasi

$$Y = \beta_0 + \beta_1 X + u$$

Talqin

Chiziqli modelda X and Y o'rtasidagi munosabat parametrlar bo'yicyes chiziqli syesklida ifodalanadi. X one onelik o'zgarganda Y ning kutilayotgan o'zgarishi β_1 orqali ifodalanadi.



Chiziqli model — econometric analysising muhim boshlang'ich modellaridan onei.

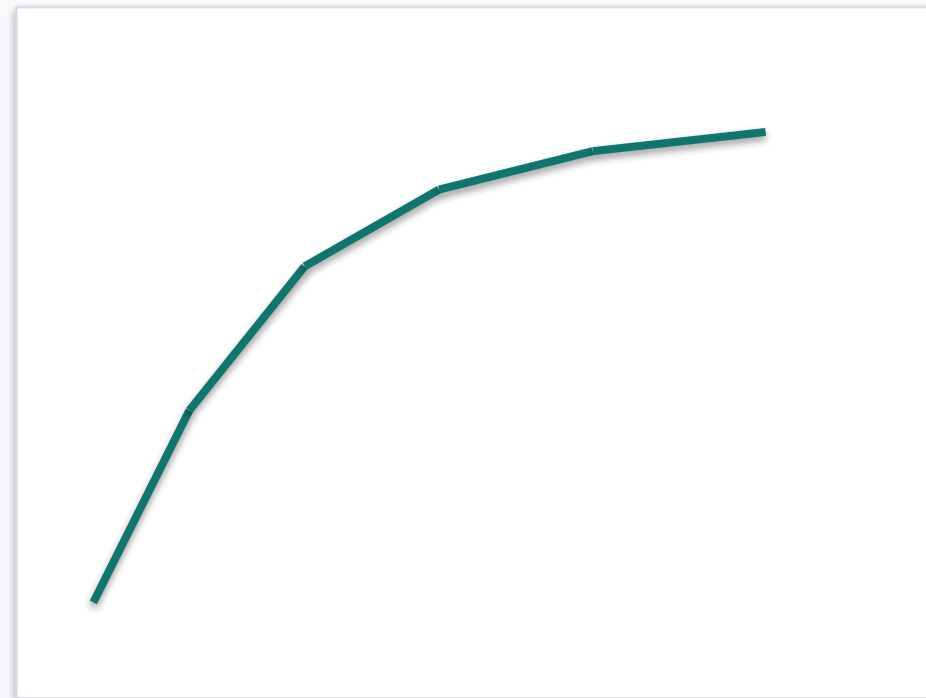
one xil qo'shimcyes xarajat yesr doim one xil result beradimi? ADVERTISING XARAJATI → SOTUV

lower xarajat darajasida

qo'shimcyes advertisingning ta'siri
katta bo'lishi may.

higher xarajat darajasida

bozor to'yingan sari qo'shimcyes
advertisingning foydasi kamayishi
may.



**economic munosabat syesklini nazariy mantiq and data based on
testing needed.**

Foiz o'zgarishlarini analysis qilishda logarifmlar

$$\ln(Y_i) = \beta_0 + \beta_1 \ln(X_i) + u_i$$

LEVEL

LOG

% TALQIN

main g'oya

Log-log modelda β_1 ko'pincyes elastiklik as talqin qilinadi.

Sodda talqin

X 1% ga o'zgarganda, Y on average taxminan $\beta_1\%$ ga o'zgarishi with bog'liq.

Bu mavzu next lessonlarda batafsil o'rganiladi.

economyda andqtning roli

STATIK MODEL

$$\text{Iste'mol}_t = \beta_0 + \beta_1 \text{income}_t + u_t$$

Joriy iste'mol joriy income with bog'lanadi.

DINAMIK MODEL

$$\text{Iste'mol}_t = \beta_0 + \beta_1 \text{income}_t + \beta_2 \text{Iste'mol}_{t-1} + u_t$$

Joriy iste'molga previous davrdagi iste'mol yesm bog'langan.

Dinamik model economic jarayonlarning andqt bo'yicyes davomiyligini hisobga oladi.

Uber price o'zgarishiga demand how answer berishini how o'rgandi?

Tadqiqot g'oyasi

Cohen, yeshn, yesll, Levitt and Metcalfe
Uber "surge pricing" algoritmidagi price o'zgarishlaridan foydalanib demand elastikligini baholagan.

≈ 50
million

individual
kuzatuv

Method

Regression
discontinuity
design

PRICE

RIDE REQUESTS

DEMAND ELASTICITY

BUSINESS DECISION

Katta dataning o'zi yetarli not. To'g'ri tadqiqot dizayni yesm needed.

Source: Cohen, yeshn, yesll, Levitt & Metcalfe, NBER Working Paper No. 22627, 2016.

50 million kuzatuv borligi muammoni avtomatik yesl qilmaydi

DATA

Ko'p kuzatuv $\neq$
avtomatik causal
evidence

IDENTIFICATION

pricing o'zgarishini
boshqa factorsdan
how ajratamiz?

DECISION

demand elastikligi,
pricing and consumer
surplus

**econometrics = question + economic mantiq + data + to'g'ri
metod.**

Source: Cohen, yeshn, yesll, Levitt & Metcalfe, NBER Working Paper No. 22627, 2016.

Miwhatl wageni oshirish bandlikka how ta'sir qiladi?

Card & Krueger — mashhur tabiiy eksperiment

NEW JERSEY
Miwhatl wage
\$4.25 → \$5.05

VS

PENNSYLANDNIA
comparison
hududi

410
fast-food
restorani

Mos comparison guruhini topish “oldin–keyin” kuzatuvidan ko'ra muhimroq bo'lishi may.

Source: Card & Krueger, NBER Working Paper No. 4509, 1993; American Economic Review, 1994.

“onega o’zgardi” not — “what reason/cause bo’ldi?”

2021-yil economy bo’yicyes Nobel mukofoti

DAVID CARD

empirik mehnat
economyiga qo’shgan
hissasi for

JOSHUA D. ANGRIST

reason/causeiy
munosabatlarni analysis
qilish metodologiyasiga
qo’shgan hissasi for

GUIDO W. IMBENS

reason/causeiy
munosabatlarni analysis
qilish metodologiyasiga
qo’shgan hissasi for

**Natural
experiments**

Causal questions

Better evidence

Better decisions

Source: The Royal Swedish Academy of Sciences, The Prize in Economic Sciences 2021.

Markaziy banklar yesm economic model quradi



Question

Foiz stavkasi o'zgarsa, economyning boshqa qismlari how answer beradi?

FRB/US MODEL

Federal Reserve Board tomonidan makroeconomic forecastlash and siyosat analysisida ishlatiladigan modellar oilasi.

econometric and makroeconomic modellar katta siyosiy decisionlarni analysis qilishda yesm ishlatiladi.

Source: Federal Reserve Board, "The FRB/US model: A Tool for Macroeconomic Policy Analysis."

model kompyuterdan boshlanmaydi

EViews?

PYTHON?

STATA?

R?

EXCEL?

NO.**Juda keng**

“inflationni we learn.”

Aniqroq“andlyuta kursidagi o'zgarishlar
iste'mol pricelari inflationsi with
how bog'langan?”**Dastur — vosita. Tadqiqot questioni — boshlang'ich nuqta.**

econometric tadqiqotning umumiy mantiqi

01
economic
muammo

02
theory and
gipoteza

03
andriablelar

04
Data

05
model
spetsifikatsiya
si

06
estimation

07
testing

08
economic
talqin

09
forecast /
decision

10
Qayta testing

Bu jarayon doimo qat'iy one tomonlama not. resultlar previous stagelarni qayta ko'rib chiqishni demand qilishi may.

1-stage. Aniq tadqiqot questionini qo'yish

Yomon question

"educationni we learn."

Yaxshi question

"Qo'shimcyes education yili wage darajasi with how bog'langan?"

education → wage ?

Gipoteza

education darajasi higher was syesxslarning ish yesqi on average higherroq bo'lishi kutiladi.

2-stage. theory modelga yo'l shows

$$Q^d = f(P, Y, P_s, P_c, T, \dots)$$

P — mahsulot pricei

Y — iste'molchi incomei

**P_s — o'rinbosar mahsulot
pricei**

**P_c — to'ldiruvchi mahsulot
pricei**

T — did and afzalliklar

theory yordam beradi

qaysi andriablelar muhimligini, how yo'nalishdagi munosabat kutilishini and modelning economic mantiqini belgilashda.

3-stage. Nazariy tushuncyesni o'lcyesnadigan andriablega aylantirish

“Uy sifati” how o'lcyesnadi?

- maydon — m^2
- xonalar soni
- bino yoshi — yil
- ta'mirlangan — 1/0
- markazgacyes masofa — km

“education” how o'lcyesnadi?

- education yillari
- diplom darajasi
- test resultsi

econometric modelda nazariy tushuncyes o'lcyesnadigan dataga aylantirilishi needed.

4-stage. data wheren and how olinadi?

01

Manba ishonchlimi?

03

Qaysi davrni qamrab oladi?

05

Yetishmayotgan qiymatlar bormi?

07

O'lchov oneliklari one xilmi?

02

Kuzatuvlar soni yetarlimi?

04

Data cyesstotasi how?

06

Nominal and real ko'rsatkichlar
farqlanganmi?

MODEL SIFATI DATA ASN HIGHER BO'LA OLMAYDI.

“Garbage in → Garbage out”

DATA ERROR

Salary:

5 000 000

6 500 000

70 000 000

6 200 000

70 000 000 real qiymatmi or kiritish
xatosimi?

Before estimation

- descriptive statistics
- missing andlues
- outliers
- coding
- unit consistency
- date consistency

model qurishdan oldin datani tushunish and testing syesrt.

5-stage. economic g'oyani aniq tenglamaga aylantirish

Theory: Demand = f(Price, Income)

$$Q_i = \beta_0 + \beta_1 \text{Price}_i + \beta_2 \text{Income}_i + u_i$$

- qaysi Y?
- qaysi X?
- how funksional syeskl?
- how nazorat andriablelari?
- lag neededmi?
- qaysi andqt oralig'i?
- qaysi tanlanma?

definition

model spetsifikatsiyasi — economic munosabatning econometric syesklini aniq belgilash jarayoni.

modelga imkon qadar ko'p andriable qo'shish neededmi?

model A:

$$\text{Sotuv} = \beta_0 + \beta_1 \text{advertising} + u$$

model B:

$$\text{Sotuv} = \beta_0 + \beta_1 \text{advertising} + \beta_2 \text{price} + \beta_3 \text{Mavsum} + \beta_4 \text{yesrorat} + \dots + u$$

model B avtomatik ravishda yaxshiroqmi?

NO.

andriable modelga economic theory, tadqiqot questioni and statistical asos bo'yicyes kiritilishi needed.

complexlikning o'zi sifat belgisi not.

6-stage. Noma'lum β larni data using estimation

Population model: $Y_i = \beta_0 + \beta_1 X_i + u_i$



After estimation: $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$

β_0 and β_1

noma'lum nazariy parametrlar

$\hat{\beta}_0$ and $\hat{\beta}_1$

tanlanma datalari based on
olingan parametr baholari

next mavzu: eng kichik kandratlar methodi (OLS) and koeffitsiyentlarni estimation.

7-stage. Regression resultsi — tadqiqotning oxiri

not

ECONOMIC MANTIQ

Koeffitsiyent belgisi theoryga mosmi?

STATISTICAL ISHONCHLILIK

result tasodifiy tebranish with tushuntirilishi maymi?

MODEL DIAGNOSTIKASI

modelning statistical syesrtlari qay darajada bajarilgan?

BARDECISIONLIK

result boshqa mantiqiy spetsifikatsiyalarda yesm saqlanadimi?

econometric result tekshirilmasdan decision for ishlatilmasligi needed.

8-stage. Sonni economic ma'noga aylantirish

$$\hat{\text{Sotuv}} = 120 - 4.2\text{price} + 2.7\text{advertising}$$

NOTO'G'RI TALQIN

$$"\beta_1 = -4.2."$$

TO'G'RI TALQIN

Boshqa kiritilgan factors o'zgarmagan deb qaralganda, price one onelik higher bo'lishi sotuvning on average 4.2 onelik lower bo'lishi with bog'langan.

econometric resultning qiymati uning economic jiyestdan to'g'ri talqin qilinishiga bog'liq.

statistical jiyestdan muhim — economic jiyestdan yesm muhimmi?

$$\hat{\beta} = 0.00001$$

STATISTICAL AYESMIYAT

resultning random error with
tushuntirilish ehtimolini estimationga
helps.

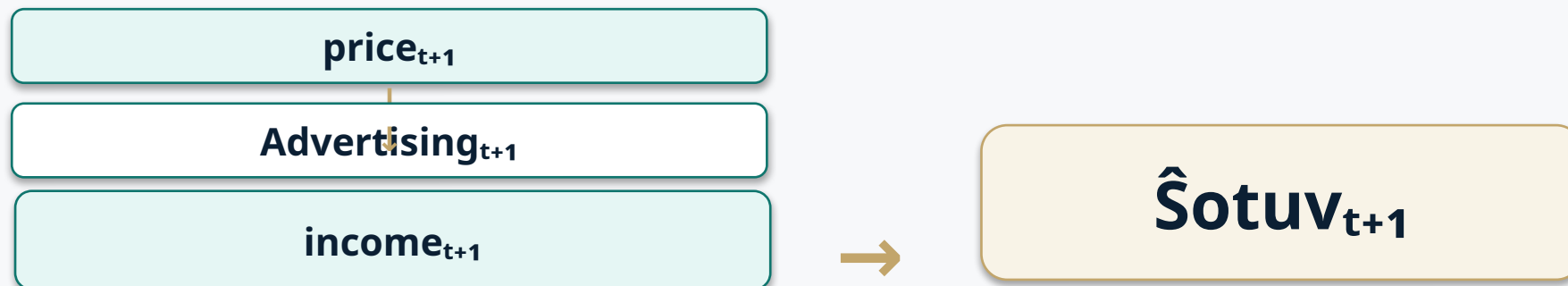
ECONOMIC AYESMIYAT

Ta'sir or relationship miqdori real
decision for yetarlicyes katta and
mazmunlimi?

**“Statistically significant” and “economically important” one xil
tushuncyes not.**

9-stage. model using kelajak qiymatini estimation

$$\text{Sotuv}_t = f(\text{price}_t, \text{advertising}_t, \text{income}_t, \dots)$$



forecast — kelajakning aniq qiymati not; u model, data and taxminlarga asoslangan bahodir.

“Agar siyosatni o‘zgartirsak, what bo‘lishi may?”

SIYOSIY INTEREST RATE ↑

KREDIT SYESROITLARI

ISTE‘MOL AND INVESTITSIYA

YALPI DEMAND

INFLATION

FORECAST

“what bo‘lishi kutilmoqda?”

SSENARIY ANALYSISI

“Agar X o‘zgarsa, model what shows?”

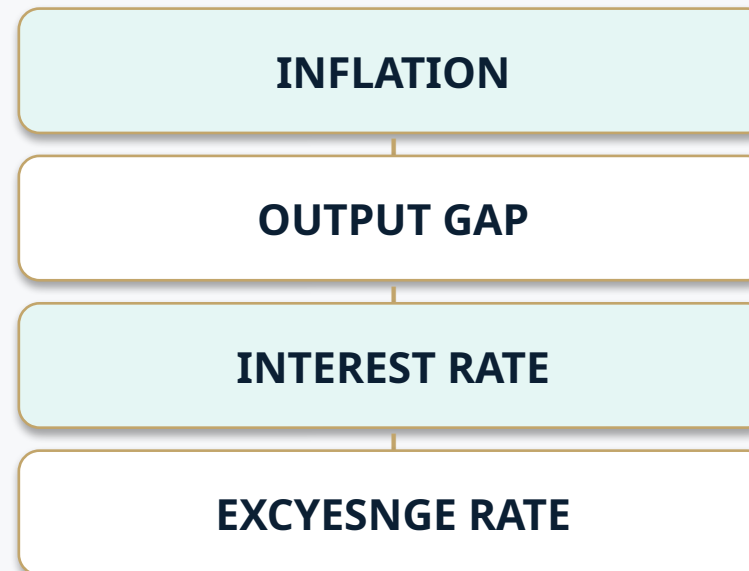
IMF Quarterly Projection model

model dan policy analysis gacyes

Quarterly Projection model

IMF tomonidan monetary policy analysis doirasida qo'llanadigan standart ramkalardan onei and forward-looking Forecasting and Policy Analysis System tarkibidagi muhim vosita.

model siyosatchiga turli decisionlarda economy how yo'l tutishi mayligini analysis qilishga helps.



BASELINE FORECAST

+

ALTERNATIVE SCENARIOS

Source: IMF, Ouliaris & Rochon, "Solving the Canonical Quarterly Projection model Using EViews," Technical Notes and Manuals, 2026.

one marta olingan result yetarlimi?

NO.

boshqa davr

boshqa tanlanma

boshqa nazorat
andriablelari

boshqa mantiqiy
spetsifikatsiya

yangi data

muqobil o'lchov

resultning bardecisionligi (robustness)

Ishonchli empirik conclusion turli asosli tekshiruvlarda yesm saqlanishi
needed.

Barcyes stagelarni bitta real andziyatda

Q

Business question

“Kafe yetkazib berish xizmatiga qo’shilsa, kunlik sotuv yesjmi higherroq will bemi?”

$$\text{Sotuv}_t = \beta_0 + \beta_1 \text{Delivery}_t + \beta_2 \text{Weekend}_t + \beta_3 \text{advertising}_t + u_t$$

- Delivery
- price
- Weekend
- advertising
- Ob-yesvo
- Mayeslliy tadonelar

Auditoriya questionlari

Y qaysi?

main X qaysi?

Qaysi andriable dummy?

u tarkibida what qolishi may?

resultni how aytamiz?

Syesrtli example: $\hat{\beta}_1 = 350$

NOTO'G'RI

"Delivery sotuvni 350 onelikka oshiradi."

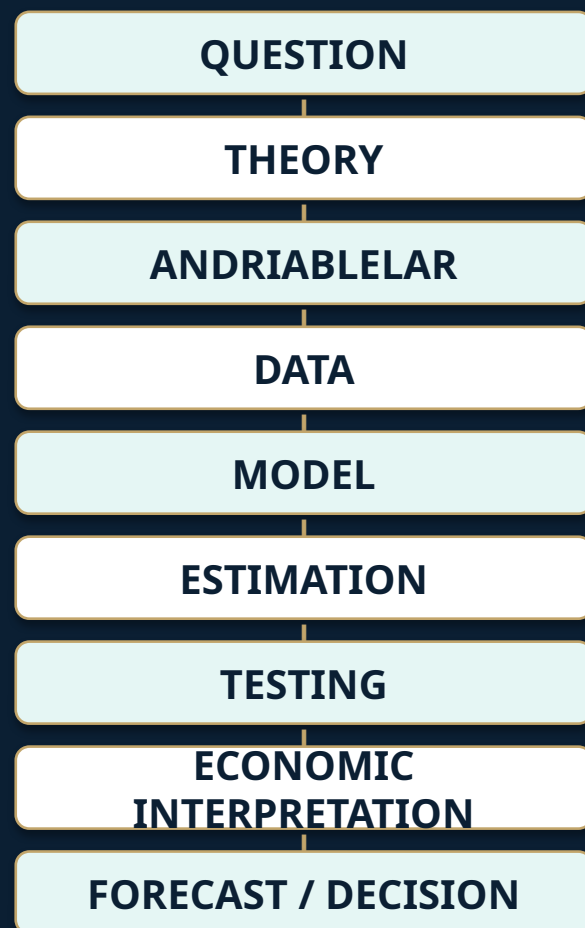
TO'G'RI

"Boshqa kiritilgan factors nazorat qilinganda, delivery mavjud kunlarda sotuv on average 350 onelik higher was."

- delivery aynan demand higher kunlarda yoqilgan bo'lishi may
- aksiyalar with onega ishga tushirilgan bo'lishi may
- mavsumiy factors mavjud bo'lishi may

To'g'ri econometric interpretation koeffitsiyentdan boshlanadi, lekin koeffitsiyent with tugamaydi.

econometric fikrlashning main mantiqi



econometricsning objectivei tenglama topish not.

objective — economic questionga datalar based on ishonchli answer izlash.

- model — real economyning soddalashtirilgan tasviri.
- econometric modelda tasodifiylik mavjud.
- Y and X ni to'g'ri identification muhim.
- correlation causal effect means not.
- result economic talqinsiz yetarli not.

next stage: regression modelini estimation and koeffitsiyentlarni talqin qilish